

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS

NANYANG TECHNOLOGICAL UNIVERSITY

MH2500

Tutorial 1, 08/2021

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We will discuss several questions from Chapter 1 in Ross' book.

Tutorial Questions

- (1) (a) How many different 7-place license plates are possible if the first 2 places are for letters and the other 5 for numbers?
(b) Repeat part (a) under the assumption that no letter or number can be repeated in a single license plate.
- (2) For years, telephone area codes in the United States and Canada consisted of a sequence of three digits. The first digit was an integer between 2 and 9, the second digit was either 0 or 1, and the third digit was any integer from 1 to 9. How many area codes were possible? How many area codes starting with a 4 were possible?
- (3) How many different letter arrangements can be made from the letters
 - (a) M i s s i s s i p p i
 - (b) a r r a n g e
- (4) If 12 people are to be divided into 3 committees of respective sizes 3, 4, and 5, how many divisions are possible?
- (5) Delegates from 10 countries, including Russia, France, England, and the United States, are to be seated in a row. How many different seating arrangements are possible if the French and English delegates are to be seated next to each other and the Russian and U.S. delegates are not to be next to each other?
- (6) From a group of n people, suppose that we want to choose a committee of k , $k \leq n$, one of whom is to be designated as chairperson.
 - (a) By focusing first on the choice of the committee and then on the choice of the chair, argue that there are $\binom{n}{k}k$ possible choices.
 - (b) By focusing first on the choice of the nonchair committee members and then on the choice of the chair, argue that there are $\binom{n}{k-1}(n-k+1)$ possible choices.
 - (c) By focusing first on the choice of the chair and then on the choice of the other committee members, argue that there are $n\binom{n-1}{k-1}$ possible choices.
 - (d) Conclude from parts (a), (b), and (c) that

$$k\binom{n}{k} = (n-k+1)\binom{n}{k-1} = n\binom{n-1}{k-1}.$$

- (7) The following identity is known as Fermats combinatorial identity:

$$\binom{n}{k} = \sum_{i=k}^n \binom{i-1}{k-1}$$

for $n \geq k$. Give a combinatorial argument (no computations are needed) to establish this identity.

- (8) Suppose that 10 fish are caught at a lake that contains 5 distinct types of fish.
- (a) How many different outcomes are possible, where an outcome specifies the numbers of caught fish of each of the 5 types?
 - (b) How many outcomes are possible when 3 of the 10 fish caught are trout?
 - (c) How many when at least 2 of the 10 are trout?

Assignment Questions

- (1) How many outcome sequences are possible when a die is rolled four times, where we say, for instance, that the outcome is 3, 4, 3, 1 if the first roll landed on 3, the second on 4, the third on 3, and the fourth on 1?
- (2) A child has 12 blocks, of which 6 are black, 4 are red, 1 is white, and 1 is blue. If the child puts the blocks in a line, how many arrangements are possible?
- (3) A student has to sell 2 books from a collection of 6 math, 7 science, and 4 economics books. How many choices are possible if
 - (a) both books are to be on the same subject?
 - (b) the books are to be on different subjects?
- (4) A psychology laboratory conducting dream research contains 3 rooms, with 2 beds in each room. If 3 sets of identical twins are to be assigned to these 6 beds so that each set of twins sleeps in different beds in the same room, how many assignments are possible?
- (5) Ten weight lifters are competing in a team weightlifting contest. Of the lifters, 3 are from the United States, 4 are from Russia, 2 are from China, and 1 is from Canada. If the scoring takes account of the countries that the lifters represent, but not their individual identities, how many different outcomes are possible from the point of view of scores? How many different outcomes correspond to results in which the United States has 1 competitor in the top three and 2 in the bottom three?
 - (a) In how many ways can 3 boys and 3 girls sit in a row?
 - (b) In how many ways can 3 boys and 3 girls sit in a row if the boys and the girls are each to sit together?
 - (c) In how many ways if only the boys must sit together?
 - (d) In how many ways if no two people of the same sex are allowed to sit together?
- (6) Give a combinatorial argument (no computations are needed) to establish the following identity:

$$\binom{n+m}{r} = \binom{n}{0}\binom{m}{r} + \binom{n}{1}\binom{m}{r-1} + \cdots + \binom{n}{r}\binom{m}{0}.$$

Then prove that

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2.$$

- (7) Consider the following combinatorial identity:

$$\sum_{k=1}^n k \binom{n}{k} = n \cdot 2^{n-1}.$$

- (a) Give a combinatorial argument for this identity by considering a set of n people and determining, in two ways, the number of possible selections of a committee of any size and a chairperson for the committee.
- (b) Give a combinatorial argument for the following identity:

$$\sum_{k=1}^n \binom{n}{k} k^2 = 2^{n-2} n(n+1).$$

(c) By using (a) and (b), prove

$$\sum_{k=1}^n \binom{n}{k} k^3 = 2^{n-3} n^2 (n+3).$$

(8) Determine the number of vectors $(x_1, \dots, x_n)$ such that each x_i is a positive integer and

$$x_1 + x_2 + \dots + x_n \leq k$$

where $k \geq n$.